

Unit 5, Lesson 10: Solving Real-World Problems on a Coordinate Plane with the Pythagorean Theorem

Benchmark

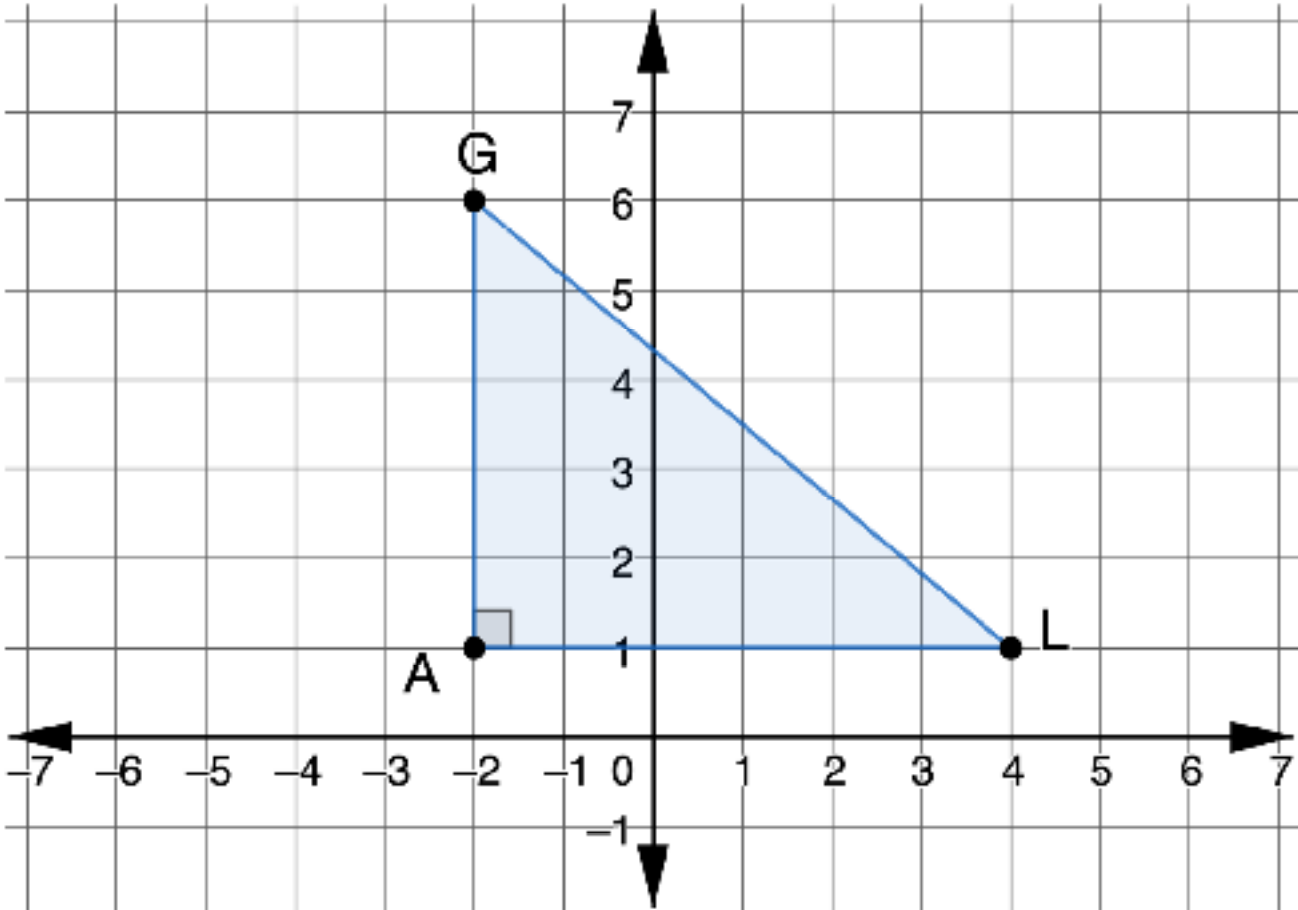
MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Learning Targets

- ☐ I can apply the Pythagorean Theorem to determine the distance between two points on a coordinate plane.
- ☐ I can apply the Pythagorean Theorem in a real-world context.

5.10.1 Warm-Up: Bell Work

Triangle GLA is shown on the coordinate plane.



1. Find the exact length of $\overline{GL}$.

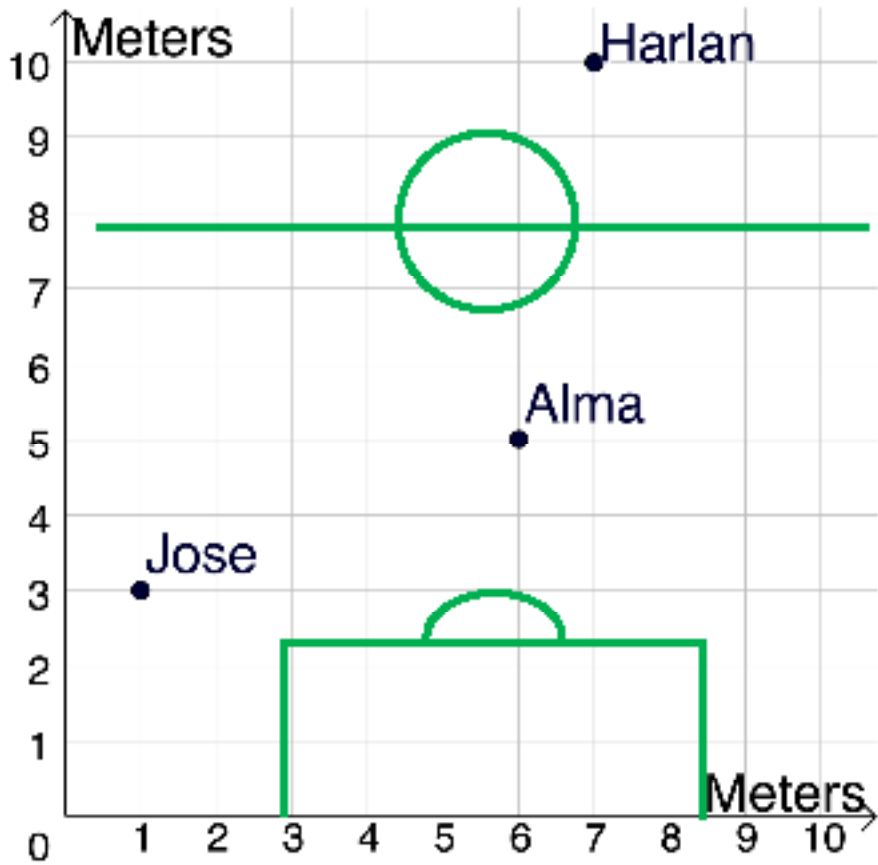
5.10.2 Collaboration: Pass the Ball!

1. Work with your partner to answer the following questions.

The figure shows the position of different soccer players during a recent game. Jose wants to pass the ball to Harlan.

He has two options.

- A. Pass the ball directly to Harlan.
- B. Pass the ball to Alma, who can pass the ball to Harlan.



- a. Which strategy, A or B, do you think will get the ball to Harlan faster?
- b. How far does the ball travel with strategy A?

c. How far does the ball travel with strategy B?

Jose and Alma pass the ball at different speeds. Jose can kick the ball forward at a rate of 2 meters per second, while Alma can kick the ball forward at a rate of 4 meters per second.

d. Calculate the time that each strategy would take.

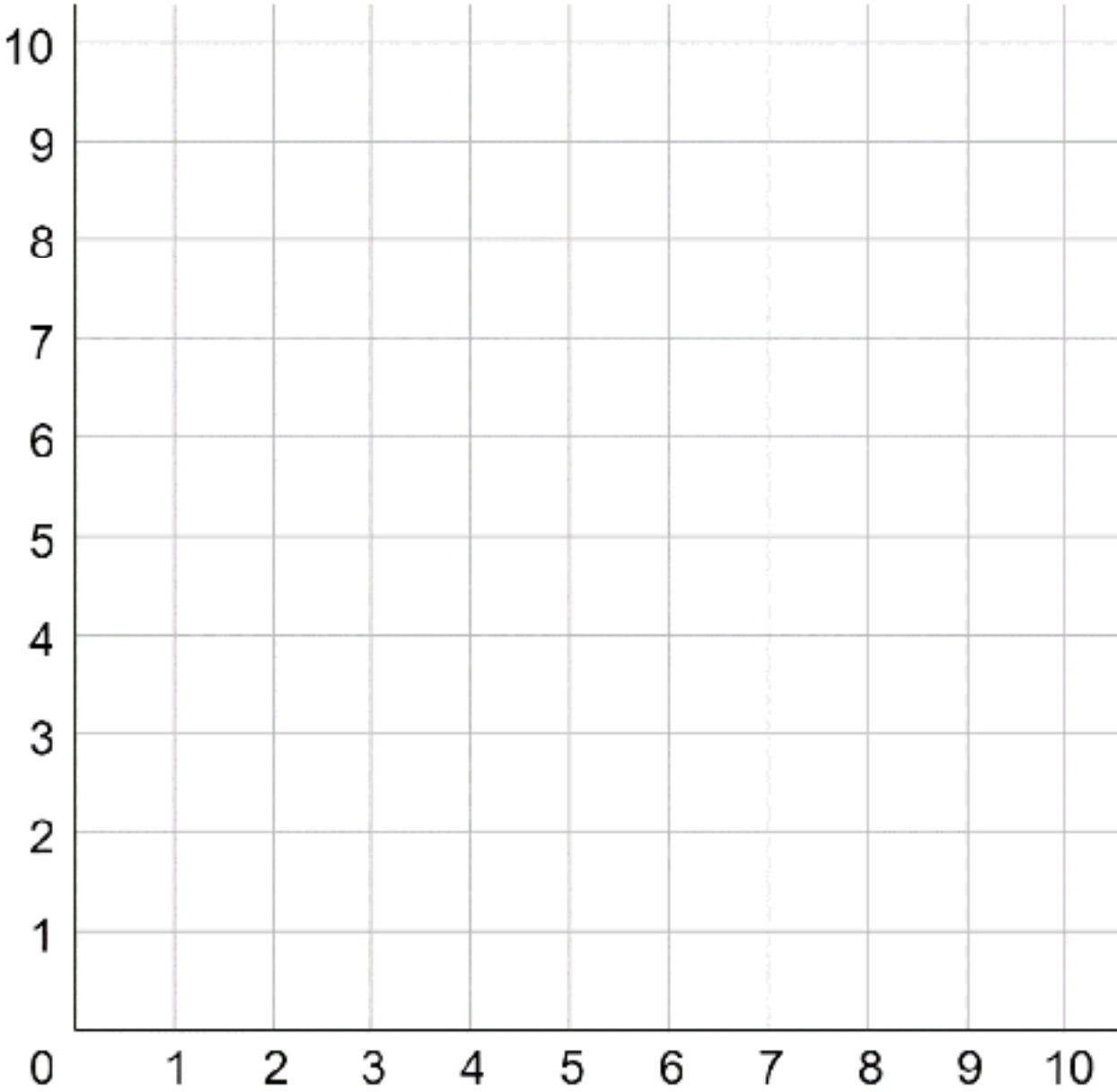
$$\frac{\text{Distance}}{\text{Rate}} = \text{Time}$$

Jose to Harlan	Jose to Alma	Alma to Harlan
Total Time for Strategy A:		Total Time for Strategy B:

e. How did the actual times for each strategy compare to your prediction?

5.10.3 Activity: Target Practice

You and your partner will attempt to guess the coordinates of a point on each other’s graphs. If the guess is incorrect, the owner of the board will calculate the distance between the two points, then share the distance with the guesser. Take turns guessing the coordinates of each other’s points and calculating the distance between the two. The first partner to guess the coordinate correctly or with the closest distance after 4 rounds, wins!



1. Plot an ordered pair on the grid (do not let your partner see it).

2. Record your partner's guesses and your guesses in the tables below.

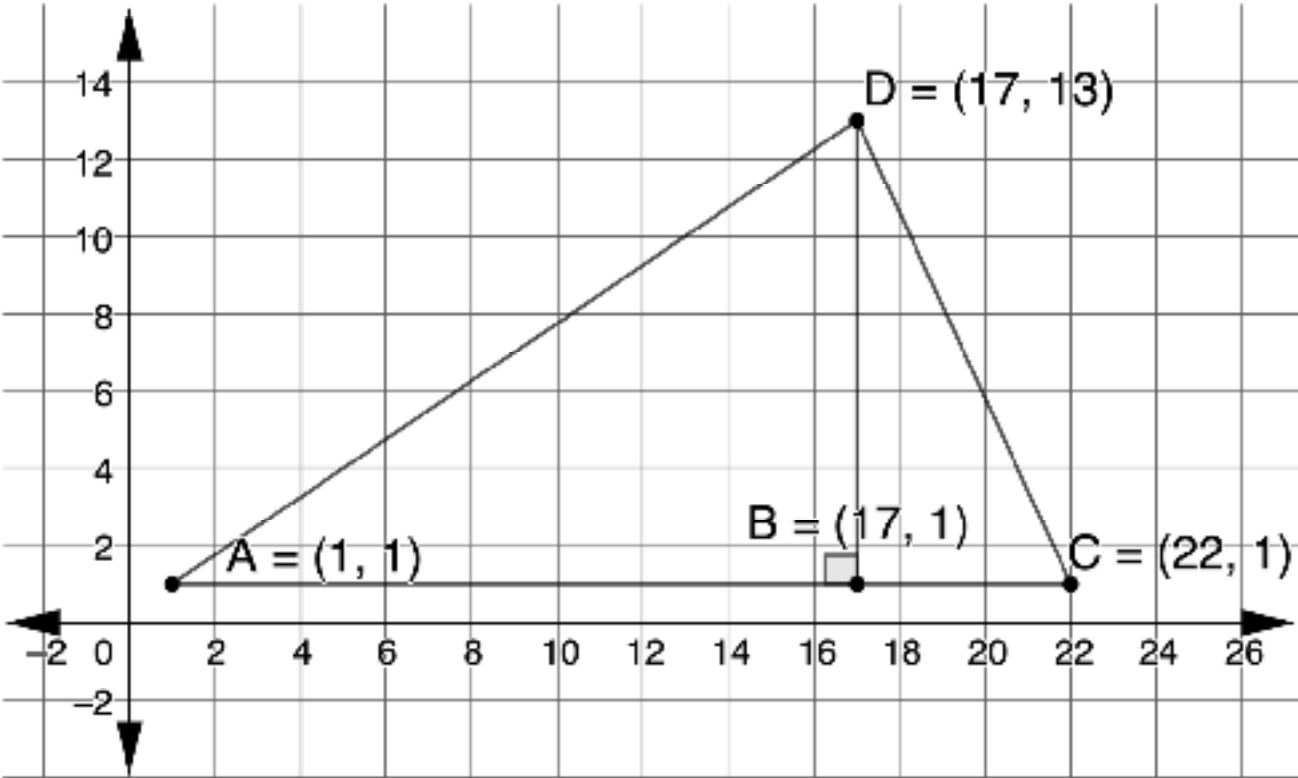
Round	Your Point	Partner's Guess	Distance Calculations
1			
2			
3			
Final Guess			

Round	Your Guess	Distance From Partner's Point
1		
2		
3		
Final Guess		

3. What strategies did you use to help determine your next guess?

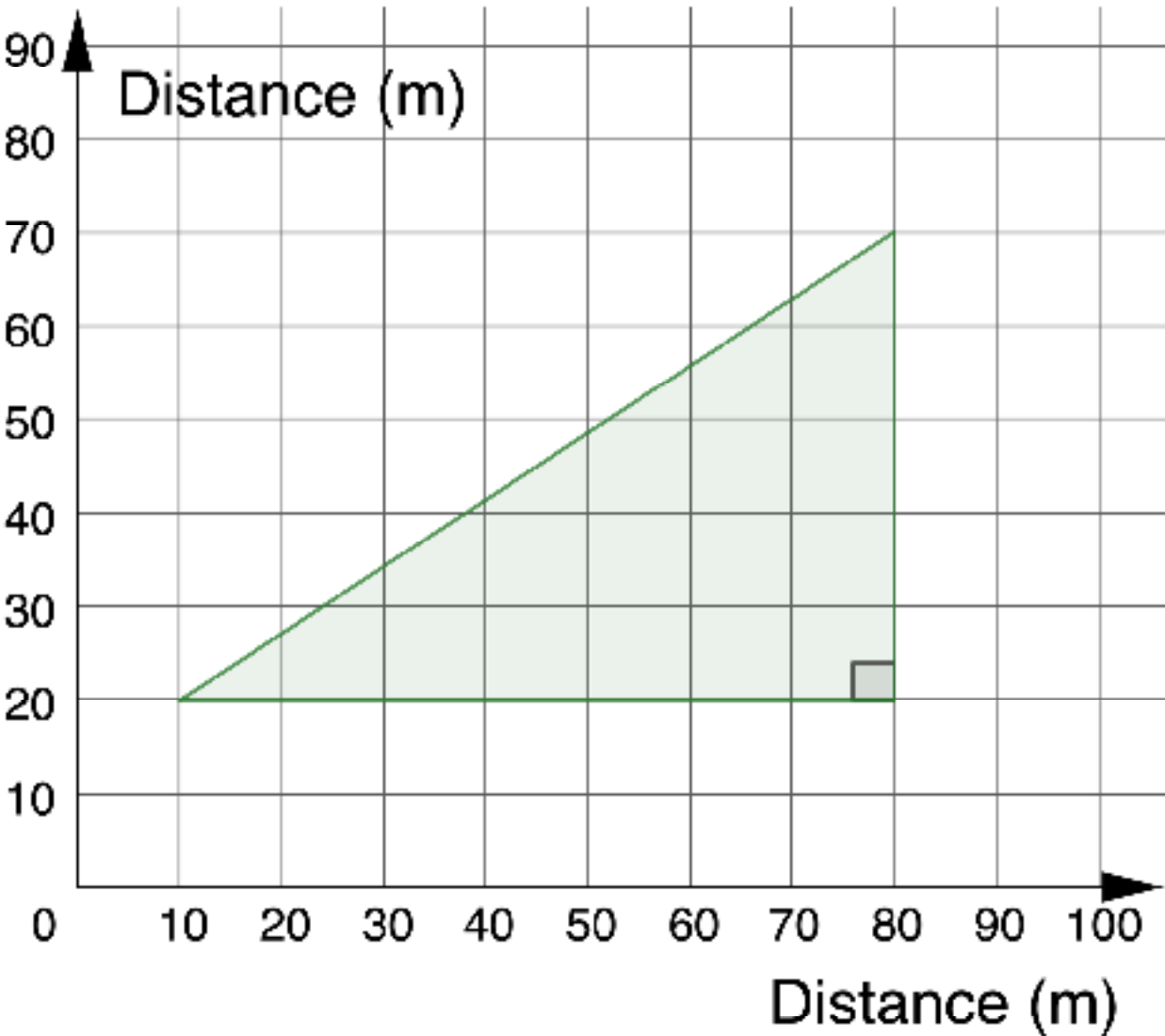
5.10.4 Your Turn!

1. Triangle ABD is shown on the coordinate plane.



Find the perimeter of Triangle ABD .

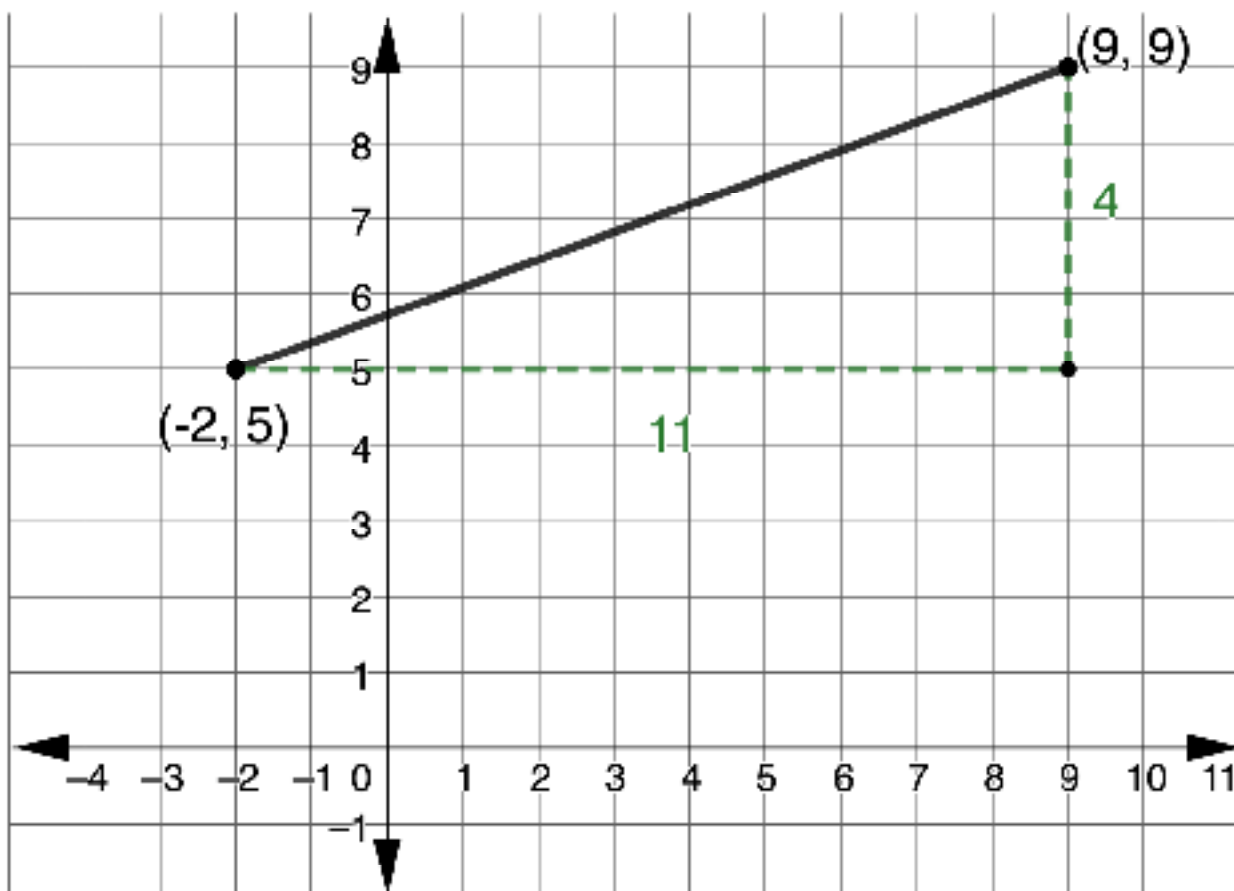
2. A walkway is constructed around a triangular field, as shown below.



How far is the walk around the field? Round your answer to the nearest tenth.

5.10.5 Wrap-Up: Error Analysis

Ellena found the distance between the points $(-2, 5)$ and $(9, 9)$. Her work is shown.



$$4^2 + b^2 = 11^2$$

$$16 + b^2 = 121$$

$$b^2 = 105$$

$$b = \sqrt{105}$$

The distance is $\sqrt{105}$ units.

1. Circle the error(s) made.
2. Write a short explanation to Ellena explaining the mistake and what she needs to remember about determining the distance between two points on a coordinate plane.